UNIVERSITI TEKNOLOGI MARA
MARKING SCHEME

COURSE	:	DATABASE DESIGN AND DEVELOPMENT
COURSE CODE	:	ICT450
EXAMINATION	:	NOVEMBER 2026 (MOCK)
TIME	:	3 HOURS

MARKING SCHEME NOTES

1. This marking scheme accompanies the ICT450 mock paper (November 2026). Answers are unofficial model answers, not official UTM marking schemes.

2. Marking points state the expected answer and the allocation of marks; every question retains its source paper and page reference.

3. Five questions x 20 marks; total = 100 marks.

4. Real-bank questions: 5. Placeholders: 0. Questions and answers are for local study use only.

5. Structure derived from 12 unique ICT450 past papers; every real question remains traceable to its source page.

FOR MARKERS' USE ONLY

This marking scheme consists of 6 printed pages.

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MARKING GUIDE

QUESTION 1 (20 marks)

Source: ICT450-2024-JUL · page [2] · bank id ICT450-2024-JUL-Q1

Reference answer

(a) Two DBMS functions for integrity and consistency. [4 marks]

- Integrity enforcement — the DBMS applies primary-key, foreign-key, domain and check constraints.
- Concurrency control — transactions are isolated so simultaneous users do not overwrite each other.
- Backup and recovery also protect consistency after failure.

(b) Four database application areas. [4 marks]

- Banking and finance.
- Airline and hotel reservations.
- University student records.
- E-commerce and online retail.
- Healthcare and hospital records.

(c) Three problems in Gerai 17's table. [6 marks]

- Missing Menu Code for Bihun Goreng — the row cannot be uniquely identified.
- Missing Price for Nasi Ayam — the total and average price cannot be calculated.
- Invalid Price “2.S0” for Laksa — the value is not numeric and violates the price domain.
- The table also mixes menu identity and price data without a clear key.

(d) Centralized versus decentralized design. [6 marks]

- Control: centralized design keeps data and administration at one site; decentralized design distributes control.
- Location: centralized data is stored centrally; decentralized data is stored at multiple sites.
- Cost/failure: centralization has lower duplication but a single point of failure; decentralization improves local availability but increases coordination cost.

Marking guide

- a) 2 marks for each DBMS function (4).
- b) 1 mark for each of four applications (4).
- c) 2 marks for each of three problems with justification (6).
- d) 2 marks for each of three differences (6).

Provenance: model_unofficial · confidence high · Model-authored from the approved question text, the supplied source assets and the ICT200/ICT450 knowledge base; no official ICT450 marking scheme was supplied.

Knowledge base basis: TOPIC 1 DATABASE CONCEPT (1).pdf; TOPIC 2 DATA MODEL (1).pdf; TOPIC 3 RELATIONAL DATA MODELstu.pdf

Assumptions / notes

- Table 1's blank Menu Code and missing/invalid Price are treated as data-entry and domain problems.

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QUESTION 2 (20 marks)

Source: ICT450-2025-FEB · page [2] · bank id ICT450-2025-FEB-Q2

Reference answer

(a) Two disadvantages of the hierarchical model. [4 marks]

- It is rigid: each child has only one parent, so many-to-many relationships cannot be represented directly.
- Navigation is pointer-based and programs depend on the physical tree structure, making change difficult.

(b) Four sources of business rules. [4 marks]

- Company policies and procedures.
- Interviews with managers and end users.
- Existing forms, reports and documents.
- Government regulations and industry standards.

(c) Three types of backup. [6 marks]

- Full backup — complete copy of the database.
- Differential backup — changes since the last full backup.
- Transaction-log backup — the transaction log for point-in-time recovery.

(d) Three mobile-database advantages in marketing. [6 marks]

- Location-based promotions can be delivered to customers near a store.
- Sales staff can access customer and product data in real time during visits.
- Campaign performance and customer responses can be captured immediately for analysis.

Marking guide

- a) 2 marks for each disadvantage (4).
- b) 1 mark for each of four sources (4).
- c) 2 marks for each backup type (6).
- d) 2 marks for each advantage (6).

Provenance: model_unofficial · confidence high · Model-authored from the approved question text, the supplied source assets and the ICT200/ICT450 knowledge base; no official ICT450 marking scheme was supplied.

Knowledge base basis: TOPIC 2 DATA MODEL (1).pdf; TOPIC 6 DATABASE DESIGN (1).pdf

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QUESTION 3 (20 marks)

Source: ICT450-2025-JUL · page [4] · bank id ICT450-2025-JUL-Q3

Reference answer

Table 3 mixes employee, department and skill data. The skill list repeats for employees with several skills, while some employees have no skill.

(a) Normal form of Table 3. [3 marks]

- The flattened table is in 1NF because each cell is atomic and the key (StaffID, SkillID) can be formed for employees with skills.
- It is not in 2NF because StaffName, Position and Department depend only on StaffID, while SkillName depends only on SkillID.
- It is not in 3NF because DeptName depends on DeptID, which is not the whole key.

(b) Keys and dependencies. [6 marks]

Flattened relation.

```
EMPLOYEE_SKILL (StaffID, StaffName, Position, DeptID, DeptName, SkillID, SkillName, CompletionDate)
```

Functional dependencies.

```
StaffID → StaffName, Position, DeptID
DeptID → DeptName
SkillID → SkillName
StaffID, SkillID → CompletionDate
```

(c) Normalise to 3NF. [11 marks]

1NF: one row per employee-skill certification; employee 302 must be kept in EMPLOYEE with no skill row.

2NF: separate employee and skill partial dependencies.

2NF relations.

```
EMPLOYEE (StaffID, StaffName, Position, DeptID, DeptName)
SKILL (SkillID, SkillName)
EMPLOYEE_SKILL (StaffID, SkillID, CompletionDate)
```

3NF: separate DeptID → DeptName.

3NF relations.

```
DEPARTMENT (DeptID, DeptName)
EMPLOYEE (StaffID, StaffName, Position, DeptID)
SKILL (SkillID, SkillName)
EMPLOYEE_SKILL (StaffID, SkillID, CompletionDate)
```

Sample employee data

StaffID	StaffName	Position	DeptID
304	Intan Syafinaz	Analyst	D1
302	Mohd Fikri Mohd Ali	Developer	D2

Marking guide

- a) 3 marks: 1NF plus 2NF/3NF reasons.
- b) 6 marks: key and dependencies.
- c) 11 marks: 1NF (2), 2NF (4), 3NF (5).

Provenance: model_unofficial · confidence high · Model-authored from the approved question text, the supplied source assets and the ICT200/ICT450 knowledge base; no official ICT450 marking scheme was supplied.

Knowledge base basis: TOPIC 5 NORMALIZATION (1).pdf

Assumptions / notes

- An employee with no skill (302) has no row in the associative table; 'Not complete yet' is treated as a NULL completion date.

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QUESTION 4 (20 marks)

Source: ICT450-2026-FEB · page [4] · bank id ICT450-2026-FEB-Q4

Reference answer

Mechanic center schema.

```
CENTER (Center_ID, Center_Name, Center_City)
MECHANIC (Mech_ID, Mech_Name, Mech_Specialty, Mech_Salary, Center_ID*)
CUSTOMER (Cust_ID, Cust_Name, Cust_CarModel)
SERVICE (Service_ID, Service_Date, Mech_ID*, Cust_ID*)
```

(a) Create the MECHANIC table. [4 marks]

```
CREATE TABLE MECHANIC (
    Mech_ID        VARCHAR(10) NOT NULL,
    Mech_Name      VARCHAR(60) NOT NULL,
    Mech_Specialty VARCHAR(30),
    Mech_Salary    DECIMAL(12,2),
    Center_ID      VARCHAR(10),
    PRIMARY KEY (Mech_ID),
    FOREIGN KEY (Center_ID) REFERENCES CENTER (Center_ID)
);
```

(b) ENGINE mechanics earning more than RM5000, sorted by salary descending. [4 marks]

```
SELECT Mech_Name, Mech_Salary
FROM MECHANIC
WHERE Mech_Specialty = 'ENGINE'
  AND Mech_Salary > 5000
ORDER BY Mech_Salary DESC;
```

(c) Different customers served at Melaka centers. [6 marks]

```
SELECT COUNT(DISTINCT s.Cust_ID) AS total_customers
FROM SERVICE s
JOIN MECHANIC m ON m.Mech_ID = s.Mech_ID
JOIN CENTER c   ON c.Center_ID = m.Center_ID
WHERE c.Center_City = 'Melaka';
```

(d) Center and mechanic with the highest salary, using a subquery. [6 marks]

```
SELECT c.Center_Name, c.Center_City, m.Mech_Name
FROM CENTER c
JOIN MECHANIC m ON m.Center_ID = c.Center_ID
WHERE m.Mech_Salary = (
    SELECT MAX(Mech_Salary)
    FROM MECHANIC
);
```

Marking guide

- a) 4 marks: columns (2), primary key (1), foreign key (1).
- b) 4 marks: specialty (1), salary filter (1), ORDER BY (2).
- c) 6 marks: three-table join (3), Melaka filter (1), COUNT DISTINCT (2).
- d) 6 marks: join (2), MAX subquery (3), selected columns (1).

Provenance: model_unofficial · confidence high · Model-authored from the approved question text, the supplied source assets and the ICT200/ICT450 knowledge base; no official ICT450 marking scheme was supplied.
Knowledge base basis: TOPIC 7 SQL.pdf

Assumptions / notes

- Mech_Salary is numeric and Center_City stores the city name.

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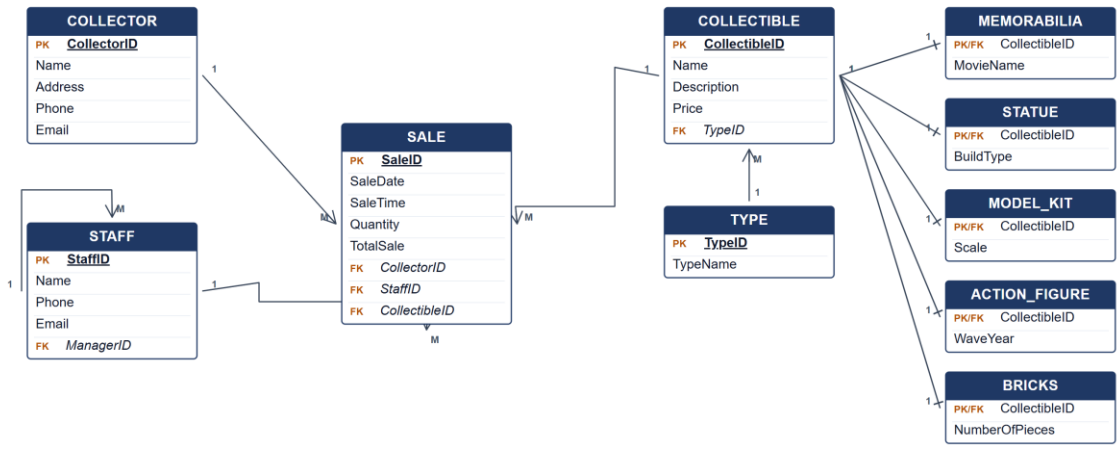
QUESTION 5 (20 marks)

Source: ICT450-2018-DEC · page [4, 5] · bank id ICT450-2018-DEC-Q5

Reference answer

The ERD separates collector, staff, sale and collectible data. SALE resolves the many-to-many purchase between collectors and collectibles, and COLLECTIBLE is specialised into the five collectible types.

(a) Construct the ERD with Crow's Foot notation. [16 marks]



Collector's Hub ERD.

(b) Two reports from the ERD. [4 marks]

- Sales by collectible type — total quantity and sales value per type, with the subtype attributes.
- Staff performance report — number and value of sales managed by each staff member, grouped by manager.

Marking guide

- a) 16 marks: collector, staff, sale and collectible entities (6); type and five subtype entities (4); relationships and cardinalities (4); keys (2).
- b) 2 marks for each report (4).

Provenance: model_unofficial · confidence medium · Model-authored from the approved question text, the supplied source assets and the ICT200/ICT450 knowledge base; no official ICT450 marking scheme was supplied.
Knowledge base basis: TOPIC 4 ERD_stu (2).pdf; TOPIC 3 RELATIONAL DATA MODELstu.pdf

Assumptions / notes

- A sale transaction records one collectible line; the manager relationship is recursive on STAFF.

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